PAPER 4

Introduction

Alzheimer's disease is a progressive neurodegenerative disorder that affects millions worldwide. Early detection is crucial for managing the disease effectively. Deep learning techniques, particularly Convolutional Neural Networks (CNNs), have demonstrated promising results in medical image analysis for Alzheimer's classification. This project aims to implement the DEMNET model, a CNN-based architecture, to classify MRI images into different stages of Alzheimer's disease using a dataset stored in a Parquet format.

Research Paper Used

- Murugan, S., Venkatesan, C., Sumithra, M. G., Gao, X-Z., Elakkiya, B., Akila, M., & Manoharan, S. "DEMNET: A Deep Learning Model for Early Diagnosis of Alzheimer Diseases and Dementia From MR Images." IEEE.

Methodology

1. **Dataset**
   - Dataset Source: Kaggle (Alzheimer MRI Disease Classification Dataset)
   - Format: Stored in Parquet format with images encoded as byte data
   - Classes: Four categories (Mild Demented, Moderate Demented, Non-Demented, Very Mild Demented)
2. **Data Preprocessing**
   - Extract images from byte-encoded format
   - Resize to (176,176) pixels
   - Apply normalization and augmentation using PyTorch's torchvision.transforms
   - Split into training (80%) and validation (20%) sets

3. **Model Architecture (DEMNET)**
    - **Convolutional Layers:** Feature extraction with 32, 64, 128, and 256 filters
    - **Batch Normalization:** Applied after each convolution to stabilize training
    - **MaxPooling Layers:** Reduce spatial dimensions and computational load
    - **Dropout Layers:** Prevent overfitting (values: 0.7, 0.5, and 0.2 at different dense layers)
    - **Fully Connected Layers:** Three dense layers followed by SoftMax for classification
4. **Training Process**
    - Loss Function: Categorical Cross-Entropy
    - Optimizer: RMSprop (learning rate: 0.001)
    - Epochs: 10
    - Batch Size: 32
    - GPU Acceleration: Training executed on NVIDIA RTX 3050 (4GB VRAM)

Evaluation Matrix

- **Accuracy:** Measures correct classifications over total samples
- **Loss Function Trend:** Helps track model convergence
- **Precision, Recall, and F1-Score:** Evaluates classification performance per class
- **Confusion Matrix:** Provides insight into class-wise prediction errors
- **ROC Curve & AUC Score:** Measures model's ability to distinguish between classes

Conclusion and Future Scope

This project successfully implemented the DEMNET model for Alzheimer's disease classification. The model demonstrated strong performance on validation data, confirming its effectiveness in distinguishing different stages of Alzheimer's. Future work may include:

- **Expanding Dataset:** Incorporating more diverse MRI scans to improve generalization.
- **Hyperparameter Tuning:** Experimenting with learning rates, batch sizes, and optimizers.
- **Model Optimization:** Using more advanced architectures like EfficientNet or Transformers.
- **Deployment:** Creating a web-based tool using Streamlit for real-time diagnosis.

This project highlights the potential of deep learning in medical diagnostics and contributes to advancements in AI-assisted healthcare.